

Changes and challenges of technical debt and its management during ongoing digital transformation

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ABSTRACT

In this paper, we look how changes and challenges in technical debt and its management that could emerge from the ongoing digital transformation. The digital transformation ensues from companies' opportunities to join various digital ecosystems and use external digital infrastructures to extend business operations and capabilities. Those external infrastructures cannot be controlled by their users, which is interesting especially from the service, systems and software development viewpoint. This will have also impact on technical debt and its management. We outline these changes and challenges with possible scenarios that could occur with technical debt during this change.

Keywords

Technical debt, technical debt management, technical debt research, digital transformation, digital infrastructures, digital platforms, digital ecosystems

1. INTRODUCTION

Digitalization [1] is a global phenomenon with a powerful impact on societies. Digitalization as a term can be defined as an integration of digital technologies by the digitization of everything that can be digitized [2]. Digitizing is the process of converting analog signals into a digital form, but digitalization on the other hand is a *sociotechnical process of applying digitizing techniques to broader social and institutional contexts that render digital technologies infrastructural* [3]. Digital technologies provide opportunities to change business models and add new revenue streams [4]. Digitalization and digital infrastructures are part of *digital transformation* that integrates digital technology into all areas of a business, which will result in fundamental changes to how businesses operate and how they deliver value to customers [5].

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Essential parts of digitalization are various digital infrastructures. Digital infrastructures [3] and digital platforms [6], found from digital ecosystems [7], are examples of ongoing digital transformation. Digital ecosystems connect infrastructure providers, companies that adopt them, and end-users who use services provided by companies [8]. Digitalization and the need for any business or public service to integrate on various digital infrastructures (such as *Google services, cloud platforms, e-health, IoT platforms, Alibaba, smart grid, and blockchain* etc.) cause changes on how software, services and systems are developed. Instead of being limited by the project length and resources, as conventionally happens, the development and deployment becomes a continuous activity. Implementing the required business changes in a separate implementation project with conventional processes would be too slow and rigid. Therefore, it can be faster and easier for companies to utilize some service or feature from existing infrastructure, rather than engaging company to a long and risky development project.

Digital infrastructures are often global and remote, which means that they cannot be controlled or managed by companies utilizing them. As a simple example, a restaurant that utilizes Facebook reservation service, or human resource department integrated to LinkedIn, do not often have any power to make changes to the service provided by the infrastructure. The infrastructures evolve continuously, and any service or product that uses it must utilize their new features and reflect their changes in real-time. When the business core is digital, all business changes and reorganizations require flexible re-development of software, systems and digital services.

Technical debt has gained recent interest in software and systems development [9]. Technical debt refers to phenomenon in software and system development, where *a badly structured technical solution or architectural design in the system, incurred by either an intentional decision or an unintentional side effect, causes omitted quality and productivity* [10]. Technical debt is an essential part of software and systems development, which is the reason the impact of digital transformation will have effect to it also. We believe that digital transformation changes how technical debt emerges and how its management needs to be changed.

In this paper, we discuss how digital transformation can change software, services and systems development, especially from the perspective of technical debt and its management. Technical debt aggregation and existence in ecosystems has been coined previously in literature [11], [12]. We want to present our view based on possible scenarios in technical debt and its management that can happen during ongoing digital transformation. More and

more technical debt will start to move in the responsibility of digital infrastructures, which are not anymore reachable by companies utilizing them. This can lead to a situation where the digital infrastructures can fully dictate how the infrastructure and its integrations must be adopted, with technical debt or without, leaving utilizing companies to just adapt and accept the situation. Eventually, this could lead to new roles and activities of technical debt management to address integrations, changes and challenges brought to software and systems development. This can be especially challenging for those that are at the edge of the digital ecosystem, just using and adapting, without any power to request changes in the digital infrastructure.

2. DIGITAL INFRASTRUCTURES

Digital infrastructures can be defined as the basic information technologies and organizational structures, along with the related services and facilities necessary for an enterprise or industry to function [3]. These infrastructures can be further defined with respect to the entity being supported or enabled as global, national, regional, industry, or corporate infrastructures [3]. Digital infrastructures exist in digital ecosystems, where there is no permanent need for centralized or distributed control or for single-role behavior [13]. In a digital ecosystem, a leadership structure may be formed (and dissolved) in response to the dynamic needs of the environment [13]. A collective of firms are inter-linked by a common interest in the prosperity of a digital technology for materializing their own product or service innovation [7].

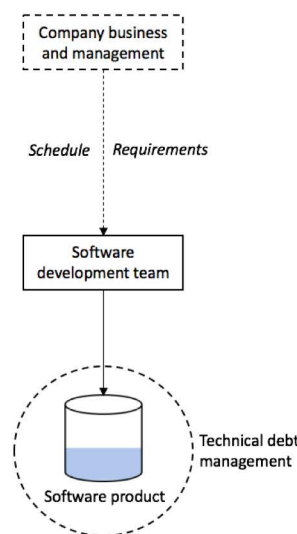
Digital infrastructures are often large, global and remote, which cannot be controlled by the people who are utilizing them. The benefit of digital infrastructures and their utilization can be relatively easy and fast integration to service or feature. However, digital infrastructure utilization is not a straightforward activity. Infrastructures have to be taken into use usually as they are, without any modifications. This emphasizes the organization's ability to adjust and adapt to technologies and adjust their organizational culture to externally created solutions. At the same time, it underlines the ability to select the most appropriate technologies and integrate them in the most effective manner. The ongoing

evolution of digital infrastructures also emphasizes the adapting organization's ability to reflect the evolution and expected changes together with own strategies. This ability to effectively utilize digital infrastructures both strategically and operationally must be enhanced and their evolution must be proactively taken into an account for novel organizational strategies and processes.

Use of external infrastructure in business is not a new trend. Companies have already used external utilities, banking services, industry-specific data interchange standards and software-as-a-service (SaaS) solutions. These solutions are based on negotiated agreements and contracts between infrastructure providers and users. However, infrastructures seem to grow bigger and global and their users have consequently less power to influence on the services and their offerings. Integration to truly external, massive and global infrastructures, e.g. to Facebook or Alibaba, does not give users much power to modify or configure the chosen services or features, which must be taken in use as they are. These types of truly external infrastructures can be especially vital for companies and their businesses that do not have an established internal IT infrastructure, because these external infrastructures can be taken into use with minimal costs.

The actors who own the infrastructure and decide its evolution based on their criteria can be called either infrastructure owners or in some cases as the focal actors, who have the power to alter the direction and evolution of digital infrastructure. However, an interesting actor that can be also found from digital ecosystems, and who we believe is at very important position from research perspective, is the *non-focal actors*, who are an ecosystem participant in the periphery of the digital ecosystem [7]. Most businesses will be non-focal from the viewpoint of digital infrastructures. Non-focal actors are the companies that will use the services of the infrastructure as they are, usually without power to make changes to infrastructure. We believe that in digital ecosystem and infrastructure related research, the perspective on non-focal actors and their software and system development, will be crucial in future.

Conventional view on
technical debt



View on **technical debt** during
digital transformation

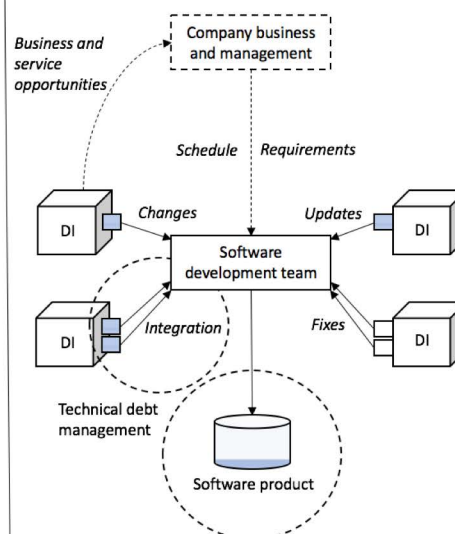


Figure 1. Views on technical debt aggregation

3. HOW DIGITAL TRANSFORMATION CHANGES TECHNICAL DEBT AND ITS MANAGEMENT?

So, what does digital transformation and technical debt has to do with each other? We describe the phenomenon and changes in Figure 1 between the conventional view and ongoing digital transformation view on technical debt. Digital transformation will have an impact on how software and systems will be developed. Rather than a more conventional way of developing software through pre-defined resources, schedules and requirements, the utilization of digital infrastructures changes the development to continuous activity, where own software solutions must be kept in sync with the changing infrastructure all the time.

The conventional view on technical debt is often seen as a need to take shortcuts or workarounds to reach time-to-market. Software development team receives schedules and requirements from company business management (often also from customer), which are used to orchestrate the software development. Sometimes the business needs of a company force software development teams to adjust the schedule and take technical debt to reach deadlines. However, in conventional view the companies often have a control over their decisions to create either shortcuts or workarounds in their own code and design, and also decide whether to pay or not pay technical debt back. Technical debt management is then organized around the technical debt that has occurred to software through own *in-house* development.

However, we believe that technical debt and its management changes when it is related to external integrations. The digital transformation forces us to use various digital infrastructures and functionality and its technical debt can move from the software product to external infrastructures. Companies can utilize services and features of digital infrastructures that include the needed business functionalities. Therefore, some of the technical debt will move from the hands of the development team to the responsibility of digital infrastructures and their developers. Much of the software development in non-focal companies is then orchestrated through integrations to infrastructures. Technical debt management now also consists the integration part from digital infrastructures, in addition to the normal technical debt management done for their own software product.

We have theorized four scenarios that could happen to technical debt and its management when non-focal companies extend their dependence on digital infrastructures.

Scenario I. Technical debt moves partly to digital infrastructures

When non-focal companies start to extend their business and capabilities with external digital infrastructures, it also increases the number of integrations to new services and capabilities. What this can do is that the code, modules, functionalities, design, and even architecture moves away from the control of the non-focal company. The needed functionalities are available through easier and faster integrations from digital infrastructures. Therefore, the emphasis of technical debt could slowly move from companies own software and systems to the other side of the integration in digital infrastructures.

Scenario II. Digital infrastructure owners will become responsible for technical debt

When non-focal companies start to utilize the opportunities emerging from the use of digital infrastructures, it will also mean that the responsibility of technical debt will slowly move towards

the digital infrastructure owners. Digital infrastructures that are relevant non-focal companies' businesses, also own the source code of the services, features and functionalities that are provided. Digital infrastructures therefore have the power to dictate the direction and evolution of the infrastructure. Owners are responsible for the development of the infrastructure and they can make decisions on their own technical debt. For example, infrastructure owners choose the parts of the infrastructure that are designed and built with highest quality and the parts that are not so strategic and can include more technical debt.

Scenario III. Non-focal companies utilizing digital infrastructures are powerless to control technical debt and its management in infrastructures

An interesting question regarding technical debt management in digital transformation is if non-focal companies can have any access or visibility to technical debt in infrastructures. When technical debt and its responsibility is moved partly to digital infrastructures, it can also mean that the effort required for technical debt management will decrease in non-focal companies. When non-focal companies just integrate their applications through APIs to infrastructures, they do not have much visibility inside the infrastructure. This creates an interesting aspect regarding technical debt *invisibility* in digital infrastructures.

Scenario IV. Technical debt management becomes more integration focused

When software and system development becomes more integration focused, it also means that technical debt management needs to adapt from the conventional way with just refactoring own source code, to refactoring of integrations to digital infrastructures. When digital infrastructures are changed, or updated, non-focal companies need to adapt their integrations. We believe that there are new roles and methods in the future for technical debt management and software engineering in general, when the importance of external infrastructures and integration to them increases.

4. FUTURE RESEARCH

The digital transformation is a business and societal change that is made possible by technology. So far it has been mostly approached from pure technical, societal or business viewpoints. There is very little research that concentrates on the interplay between these areas, especially from the service, systems or software construction viewpoints, where the capability of transformation is made possible. This interplay is also relevant to technical debt and its management and it has not been discussed widely. We see that this research opening has a big potential to produce important results for both academia and industry.

We have some open questions regarding technical debt and its management in future that could be interesting future research topics:

- How visible is technical debt in digital ecosystem and infrastructures, especially from the viewpoint of a non-focal company?
- Do non-focal companies need new practices and methods when they extend the use of external digital infrastructures?
- What is the power of a non-focal company when deciding about technical debt and its management in digital infrastructures?
- Should digital infrastructures be more open regarding their development and technical debt?

- Do digital infrastructures need to be open-source that non-focal companies have more knowledge regarding the infrastructure?
- How much business and development should non-focal companies rely on digital infrastructures?
- Will technical debt management disappear in non-focal companies in future?
- How do non-focal companies understand technical debt in their infrastructure integrations?

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